Acquiring exact phase representations from dense quadratic data

Unknown symmetric diagonal completions, sparse errors, and exact counting

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Abstract

We give a deterministic representation compiler for a raw alternating binary matrix. Suppose its off-diagonal entries agree, except for at most M unordered pairs, with an unknown symmetric matrix of rank at most r. Neither that matrix, its diagonal, its column space, nor the error locations are supplied. The compiler deletes $O(\sqrt{rM} + r)$ original coordinates and acquires an exact symmetric rank-at-most-r completion of the remaining matrix, with at most M off-diagonal errors. Its time is subexponential in the original dimension whenever $r = o(n)$ and $rM = o(n^2)$; its space is polynomial. It acquires a column code, deletes supports of short codewords until decoding is unique, and independently validates the result.

A fourth-root quadratic refinement converts the acquired completion into at most 2^r exactly weighted unary modifications. Combined with the previous constructive local-phase theorem, conditional on the pinned OpenAI counting manuscript, this computes a dense quadratic phase perturbed by m fixed-locality phases in $\mathrm{poly}(L+n)2^{o(n)}$ time and polynomial space if, additionally, $\log(2 + (m+M)/n) = o(\log n)$. In particular, $M = O(n)$ admits every $r = o(n)$. The acquisition theorem itself is unconditional. All representation, arithmetic, reconstruction, and description-size costs are charged. Executable finite validators check the new compiler and exact identities; they do not implement the asymptotic replacement-form solver. Prior exact and approximate matrix algorithms are compared, including the already-solved error-free diagonal case.

1 Input model and the two results

All matrix operations until explicitly stated otherwise are over $\mathbb{F}_2$. A symmetric matrix is *alternating* if its diagonal is zero. Write $\mathrm{off}(D) = D + \mathrm{diag}(D_{11}, \dots, D_{nn})$. For an alternating matrix S, let $e(S) = |\{(i,j) : i < j,\ S_{ij} = 1\}|$, counting unordered edges once. For an alternating B, put

$$q_B(x) = \sum_{i<j} B_{ij} x_i x_j \pmod 2.$$

The matrix is explicitly given, with at most n^2 entries. In the phase problem, L includes this matrix and every index and entry of every local phase table. Polynomial input-reading and matrix costs are charged separately from the exponent in n.

Theorem 1.1 (Acquisition from a raw matrix)**.** *Let B be an alternating n by n matrix, $n \geq 1$. Let $0 \leq r \leq n$, $0 \leq M \leq \binom{n}{2}$, and let $1 \leq \delta \leq n$ be an integer. Define*

$$A_n(t) = \sum_{j=0}^{\min(n,\lfloor t \rfloor)} \binom{n}{j}, \qquad s_0 = \min\left(n, \left\lfloor \frac{2M}{\delta} \right\rfloor + r\delta\right), \quad s = \min\left(n, \left\lfloor \frac{2M}{\delta} \right\rfloor + 3r\delta\right).$$

There is a deterministic algorithm with time

$$\mathrm{poly}(n) A_n(r) A_n(s_0) 2^r \tag{1}$$

and polynomial space which returns either failure or a verified representation

$$R = [n] \setminus E, \quad B[R,R] = \operatorname{off}(C) + S', \quad C = C^\top, \quad \operatorname{rank} C \le r, \quad e(S') \le M, \quad |E| \le s.$$

If there exists a symmetric matrix D such that

$$\operatorname{rank} D \le r, \qquad B = \operatorname{off}(D) + S, \qquad e(S) \le M,$$

the algorithm necessarily returns a representation. The matrices D, S and their existence witness are not inputs.

This is an exact representation search, not a decision procedure for the original full-matrix rank-editing promise: an accepted retained representation can exist even when that promise is false. Its identity is always checked. No incoherence, bound on the original maximum error degree, or graph decomposition is assumed.

For $r > 0$, choose

$$\delta = \max\left(1, \left\lceil \sqrt{2M/(3r)} \right\rceil\right). \tag{2}$$

Then $s \le 2\sqrt{6rM} + 3r$. For $r = 0$, directly test $e(B) \le M$ and take $C = 0, E = \varnothing$. The usual binomial-sum bound gives

$$\log_2 A_n(t) = O(t \log_2(en/t)) \quad (1 \le t \le n/2).$$

Thus (1) is $2^{o(n)} \operatorname{poly}(n)$ if $r = o(n)$ and $rM = o(n^2)$. Enumerating ordered tuples and using n^r instead would lose this whole parameter regime.

Theorem 1.2 (Dense quadratic backgrounds with acquired structure)**.** *Fix integers $k, b \ge 1$. Let ζ be a primitive 2^b-th root of unity. The input consists of B as above and m explicit tables $f_j : \{0,1\}^{I_j} \to \mathbb{Z}/2^b\mathbb{Z}$, with $|I_j| \le k$. Let*

$$Z = \sum_{x \in \{0,1\}^n} (-1)^{q_B(x)} \zeta^{\sum_{j=1}^m f_j(x|_{I_j})}.$$

Conditional on the local-phase theorem in Section 5, and hence on the specified replacement-form results in [1], there is an exact deterministic algorithm of bit time $\operatorname{poly}_{k,b}(L+n)2^{o(n)}$ and polynomial space on every family admitting the promise of Theorem 1.1 with

$$r = o(n), \qquad rM = o(n^2), \qquad \log(2 + (m+M)/n) = o(\log n). \tag{3}$$

The rank and error budgets can be discovered by dovetailing; the representation and diagonal are never supplied. In particular, $M = O(n)$ permits every $r = o(n)$.

This does not give a subexponential algorithm for an arbitrary dense quadratic background plus nonlinear phases. The favorable guarantee depends on an intrinsic rank-and-sparsity property, but its useful representation is constructed and checked.

2 Finding clean information without knowing the errors

Lemma 2.1 (An unknown clean-column witness)**.** *Under the promise of Theorem 1.1, there are sets $I, E_0 \subseteq [n]$ with $|I| \le r$, $|E_0| \le s_0$, such that for $R_0 = [n] \setminus E_0$,*

$$B[R_0, I] = D[R_0, I], \qquad \operatorname{col}(D[R_0, R_0]) \subseteq \operatorname{col}(B[R_0, I]).$$

Every column of $(B + D)[R_0, R_0]$ has weight at most δ.

Proof. In the hidden error graph let $H = \{i : \deg_S(i) \geq \delta\}$. Its degree sum is at most $2M$, so $|H| \leq \lfloor 2M/\delta \rfloor$. Put $V = [n] \setminus H$. Choose column indices $I \subseteq V$ forming a basis for $\mathrm{col}(D[V, V])$; there are at most r. Let J be the union of the S-neighbors of I within V and set $E_0 = H \cup J \cup I$. Every vertex in V has error degree at most $\delta - 1$, so $|J \cup I| \leq r(\delta - 1) + r = r\delta$. There are no off-diagonal errors between R_0 and I, and deleting I removes the unknown diagonal discrepancy from these columns. Restricting the basis relations on V proves the column-space inclusion. Remaining off-diagonal errors have column weight at most $\delta - 1$; the unknown diagonal adds at most one discrepancy per column. □

The basis indices in I may lie in the deleted set; this construction deliberately deletes them. The algorithm must keep reading the corresponding *external columns of the original matrix*. Requiring the basis indices to survive is invalid. Also the acquired column code may be a proper supercode of the true retained column space, because principal restriction can reduce the latter rank. Equality is not required.

3 Short codewords turn erasures into unique decoding

Enumerate all unordered $I \subseteq [n]$ with $|I| \leq r$ and $E_0 \subseteq [n]$ with $|E_0| \leq s_0$. For each pair, start with $R = [n] \setminus E_0$ and compute independent generators for $\mathcal{C}_R = \mathrm{col}(B[R, I]) \subseteq \mathbb{F}_2^R$. This step sees only B, I, E_0 and never accesses the hidden matrices used in Lemma 2.1.

Lemma 3.1 (Distance by deleting supports)**.** *By deleting at most $2r\delta$ further coordinates, one can make every nonzero word of the restricted code have weight greater than 2δ. This takes $\mathrm{poly}(n)2^r$ time and polynomial space. Any sequence of qualifying deletions works.*

Proof. Enumerate codewords using the current independent generators. If a nonzero word has weight at most 2δ, delete its support and restrict the generators to the surviving coordinates. This restriction map has a nonzero kernel, namely the chosen word, so its code dimension strictly decreases. Recompute a basis and repeat. There are at most r rounds, each deleting at most 2δ coordinates. At most 2^r words are scanned per round. Generate them one at a time; storing all words would violate the polynomial-space claim. □

For the witness in Lemma 2.1, inclusion of the true remaining column space in the code survives every restriction. Each observed remaining column is still within distance δ of its true column. At termination the code has minimum nonzero distance greater than 2δ, so that true column is its unique codeword within distance δ.

Decode each column of $B[R, R]$ by scanning the codewords. Reject if any column has no nearby word. Let C be the decoded matrix. Independently verify symmetry, rank at most r, and that $S' = B[R, R] + \mathrm{off}(C)$ has at most M unordered edges. The empty code is allowed; its only word is zero. A wrong seed can fail these checks or give a different valid representation, but cannot give a false accepted identity.

The correct witness decodes to $D[R, R]$ and passes. The deletion bound is $|E| \leq \min(n, s_0 + 2r\delta) = s$. Each candidate takes $\mathrm{poly}(n)2^r$ time and polynomial space, including decoding, Gaussian elimination, and verification. Multiplying by the two unordered-subset counts proves Theorem 1.1. No $2^{\Theta(r^2)}$ search over inner symmetric coefficient matrices is present.

4 The diagonal becomes unary fourth-root phases

For even n the alternating matrix $J_n + I_n$ has full rank, but the symmetric completion J_n has rank one. Thus allowing the unknown diagonal broadens the former alternating-rank promise. Diagonal completion alone is established machinery; Section 8 states the relevant prior algorithm.

Let the acquired code have $d \le r$ independent columns, in a full-column-rank matrix $U \in \mathbb{F}_2^{R\times d}$. Its space contains $\mathrm{col}(C)$. Obtain a left inverse L_U by Gaussian elimination and set $K = L_U C L_U^\top$. Then $K = K^\top$ and $C = UKU^\top$. These identities are directly checked, including when $\operatorname{rank} C < d$.

For a symmetric binary matrix T, define the $\mathbb{Z}/4\mathbb{Z}$-valued quadratic refinement

$$Q_T(z) = \sum_j T_{jj} z_j + 2\sum_{j<\ell} T_{j\ell} z_j z_\ell \pmod 4. \tag{4}$$

Its polar form under Boolean addition is

$$Q_T(x+y) - Q_T(x) - Q_T(y) = 2x^\top T y \pmod 4.$$

Lemma 4.1 (Exact correction under substitution)**.** *Let u_i be row i of U. Define $\ell_i = (C_{ii} - Q_K(u_i))/2 \pmod 2$. The numerator is even and*

$$Q_C(y) = Q_K(U^\top y) + 2\ell^\top y \pmod 4.$$

Consequently

$$(-1)^{q_{\mathrm{off}(C)}(y)} = i^{Q_K(U^\top y)} (-1)^{\ell^\top y} \prod_{j\in R} i^{-C_{jj} y_j}. \tag{5}$$

Proof. The two refinements have the same polar form because $C = UKU^\top$. Their difference is a homomorphism from the elementary abelian 2-group $\mathbb{F}_2^R$ to $\mathbb{Z}/4\mathbb{Z}$, so its values are in $\{0, 2\}$. Evaluation on coordinate vectors gives the correction. Equivalently $Q_K(u_i) \bmod 2 = u_i^\top K u_i = C_{ii}$, which verifies the division by two. Subtracting the diagonal term from (4) proves (5). □

Even for an alternating completion, Boolean substitution need not preserve the quadratic polynomial without a unary correction. For rows of U equal to $(1,1), (1,0), (0,1)$ and $K = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the substituted binary quadratic contains an extra y_1. Dropping this correction changes exact answers.

For $a \in \mathbb{F}_2^d$ put $G_K(a) = \sum_{z\in\mathbb{F}_2^d} i^{Q_K(z)} (-1)^{a^\top z} \in \mathbb{Z}[i]$. Boolean Fourier inversion gives

$$i^{Q_K(U^\top y)} = 2^{-d} \sum_{a\in\mathbb{F}_2^d} G_K(a) (-1)^{(Ua)^\top y}. \tag{6}$$

This replaces the completion by at most 2^r exactly weighted families of unary phases. Recompute and stream the coefficients; do not retain an exponentially large Fourier table.

Lemma 4.2 (Polynomial-cost coefficients)**.** *Each $G_K(a)$ can be computed in polynomial time and space using Gaussian integers with $O(d)$ bits.*

Proof. Write the exponent modulo four as $\sum_j \alpha_j z_j + 2\sum_{j<\ell} \beta_{j\ell} z_j z_\ell$, where $\alpha_j = K_{jj} + 2a_j$. If α_j is odd, let $L(z) = \bigoplus_{\ell\neq j:\beta_{j\ell}=1} z_\ell$. Then

$$1 + i^{\alpha_j} (-1)^{L(z)} = (1 + i^{\alpha_j}) i^{-\alpha_j L(z)}.$$

Modulo four, a Boolean parity is the sum of its variables minus twice their pairwise products. The remaining exponent stays quadratic: subtract α_j from each neighbor's unary coefficient and toggle each pair of neighbors' quadratic coefficients. Multiply an accumulated Gaussian integer by $1 + i^{\alpha_j}$ and eliminate z_j. After at most d steps, all unary coefficients are even. The remaining sum is an ordinary binary quadratic Gauss sum, evaluated by eliminating coupled variable pairs and checking the remaining linear form. This takes polynomially many field and integer operations. All partial factors and sums have $O(d)$ bits. □

5 The constructive local-phase dependency

The only non-elementary counting dependency is the companion local-phase construction [2]. It applies to u tables of fixed locality κ and fixed 2-power root order on n independent Boolean inputs, with unrestricted interaction graph. Its exact phase sum takes $\mathrm{poly}_{\kappa,b}(L+n)2^{F(n,u)}$ time and polynomial space, where $F(n,u)=o(n)$ if $\log(2+u/n)=o(\log n)$. More explicitly, set

$$h=\log_2(2+\kappa u/n)+O_{\kappa,b}(\log\log n),\qquad a=h/\log_2 n.$$

In the useful range $a\to 0$, one may take

$$F(n,u)=O_{\kappa,b}\Big(n\sqrt{a}\,[\log_2(2/a)]^{3/2}\Big). \tag{7}$$

Outside its guarded useful range the solver enumerates original inputs. The companion proof gives explicit guards and parameters.

This dependency does not assume small shared summaries. Multilinearly interpolate each phase-valued table ζ^{f_j} to G_j over cyclotomic integers. For $\pi=\zeta-1$, each coefficient of G_j-1 belongs to (π). At root order 2^b, put $d_0=2^{b-1}$; then $\pi^{d_0}\in 2\mathbb{Z}[\zeta]$. The retained product

$$\prod_j(1+V(G_j-1))\quad\text{in}\quad\mathbb{Q}(\zeta)[V]/(V^{d_0(n+2)})$$

has incidence degree $O_{\kappa,b}(n)$ independently of u, and omitted orders vanish modulo 2^{n+2}. Each coordinate of the exact original sum lies between -2^n and 2^n, so those residues determine it.

High-incidence variables are branched on at cost $2^{o(n)}$; remaining incidence is $n^{o(1)}$. Lemma 4.7 of [1] manufactures occurrence-specific replacement forms, variable groups, and shared integer measurement matrices. Proposition 3.4 reconstructs the retained polynomial from its $O(n)$ incidence-degree budget. Density-sensitive choices give (7); rational bit costs are controlled as required in Appendix B of the source. Apply the coefficient-sum map in V only *after* exact reconstruction: setting $V=1$ is not a homomorphism on this truncated algebra.

The new compiler therefore does not invoke an unspecified arbitrary-weight sum oracle or divide modular residues. It obtains every exact inner cyclotomic integer before the divisions below.

6 Original-input reconstruction and complete costs

Apply Theorem 1.1. For each assignment e to E, write $x=(e,y)$ with y ranging over the remaining original variables. Dense quadratic interactions incident to E become a known constant and unary phases computed from B. Restrict every original table by the same assignment; locality cannot increase. The sparse matrix S' adds at most M quadratic tables.

Use (5) and (6) for C. For each $a\in\mathbb{F}_2^d$, the inner sum contains the restricted original tables, the M sparse quadratic tables, and at most $|R|$ unary tables, combining all diagonal, substitution, character, and branch corrections. Set $b'=\max(b,2)$ and embed the original phases into the $2^{b'}$-th roots. Locality is fixed at $\kappa=\max(k,2)$ and the root order is fixed.

To use the original n uniformly in the quantitative bound, pad each inner problem with $|E|$ unused Boolean variables. Let $T_{e,a}$ be its *exact* sum, including the branch's constant phase. It is $2^{|E|}$ times the unpadded sum, so

$$Z=\sum_{e\in\mathbb{F}_2^E}2^{-(d+|E|)}\sum_{a\in\mathbb{F}_2^d}G_K(a)T_{e,a}. \tag{8}$$

The unpadded finite validator instead divides by 2^d. Assignments to E are disjoint original-input branches, with no averaging over them.

Each inner coordinate is bounded by 2^n and reconstructed with the original precision $n+2$ before forming a numerator. The Gaussian coefficients have coordinates bounded by 2^d. A

streamed numerator has $O(n+r)$ bits per coordinate. The denominator in (8) divides every coordinate exactly in the fixed integral cyclotomic basis. Only then is the quotient added to the outer accumulator. Divisibility can be checked.

Writing $u = m + M + O(n)$, the full bound is

$$\mathrm{poly}_{k,b}(L+n)\left[A_n(r)A_n(s_0)2^r + 2^{s+r+F(n,u)}\right]. \tag{9}$$

Acquisition is paid once, so it is added to the counter work. The coefficient computations, table transformations, and exact arithmetic are included. Input storage, polynomially many matrices, one branch, one codeword, one Gaussian coefficient, one inner-solver state, and exact accumulators suffice. This proves polynomial space and Theorem 1.2.

No supplied favorable budgets. Dovetail the full acquisition-and-counting pipelines over $r \in [0,n]$, $M \in [0,\binom{n}{2}]$, $\delta \in [1,n]$, and direct enumeration. There are polynomially many polynomial-space states. All accepted outputs are exact, and the first completed answer takes at most polynomial overhead relative to the best successful budget. An uncontrolled sequential search would not prove this claim.

7 A program-level consequence with all states charged

Consider a Boolean straight-line program of description size L with n independent inputs, arbitrary affine operations over $\mathbb{F}_2$, t two-input AND operations, and one requested output bit. Eliminate affine intermediate states by symbolic linear propagation. Introduce a formal result bit u_j for each AND, so its operands become computed affine forms $a_j(x,u_{<j}), b_j(x,u_{<j})$. The output is an affine form $o(x,u)$. This costs polynomial time in $L+n+t$ using vectors of length $n+t+1$, without enumerating input assignments.

Introduce operand bits v_j, w_j, character bits $\alpha_j, \beta_j, \lambda_j$, and one output character μ. For a requested value τ, character orthogonality gives

$$\#\{x : o(x) = \tau\} = 2^{-(3t+1)} \sum_{x,u,v,w,\alpha,\beta,\lambda,\mu} (-1)^{P_{\mathrm{prog}}}, \tag{10}$$

$$P_{\mathrm{prog}} = \sum_{j=1}^{t} [\alpha_j(v_j + a_j) + \beta_j(w_j + b_j) + \lambda_j(u_j + v_j w_j)] + \mu(o+\tau).$$

There are exactly $N = n + 6t + 1$ free phase variables and exactly t cubic monomials $\lambda_j v_j w_j$. Everything else is quadratic or unary. Construct the raw quadratic matrix by expanding the computed affine forms, in polynomial time. Each character enforces one constraint with a factor two; acyclicity makes all gate and operand bits unique for each x. This proves the exact normalization.

If $t = O(n)$ and this *computed* N-dimensional matrix admits the conditions of Theorem 1.2, the requested count takes $\mathrm{poly}(L+n)2^{o(n)}$ time and polynomial space, even with many more affine intermediate states than inputs. The final division by 2^{3t+1} occurs after the exact phase sum. Those conditions are not guaranteed for every program. The useful representation is acquired from the raw computed matrix, and the cost is measured against the original inputs.

8 Comparison, decisive failures, and finite execution

A genuine exact prior competitor. Fomin, Golovach, and Panolan [3], Theorem 6, solve full binary GF(2) rank-r approximation within k_0 changed entries in time $2^{O(r\sqrt{k_0 \log(rk_0)})}n^2$. For an alternating completion $k_0 = 2M$; with an unknown diagonal one can use $k_0 = 2M + n$. Their output need not be symmetric, but a polynomial support-cover repair deletes $O(\sqrt{rk_0} + r)$

coordinates and makes it symmetric. The retained residual has at most $k_0/2$ unordered off-diagonal errors, still $O(M+n)$. The accompanying audit proves and executes this repair.

For $M = O(n)$ the displayed exact bound guarantees subexponential acquisition, for example, when $r = o(\sqrt{n/\log n})$. It does not imply the whole $r = o(n)$ regime of (9). Our output task is weaker than full-matrix editing because it allows an acquired coordinate deletion. This is not an unqualified improvement to their full rigidity algorithm or a claim of first priority.

The GF(2) bicriteria algorithm of Velingker et al. [4] has time $2^{\mathrm{poly}(r/\varepsilon)}\,\mathrm{poly}(n)$ and output rank $O(r\log n/\varepsilon)$. Its approximation error can be retained as exact sparse phases; the label approximate does not disqualify it. Its stated dependencies do not supply the full sublinear-rank guarantee here. Basis-column and codeword enumeration also have relevant ancestry in incidence-matrix recovery [6]; that result uses a different residual class and $(mn)^{O(r)}$ time.

The error-free diagonal case is already solved. Kogan [5], Theorem 2 and Remark 3, constructs a diagonal completion of rank at most twice the minimum in $O(n^4)$ time. It already suffices for subexponential latent expansion when $M = 0$ and the minimum rank is $o(n)$. That algorithm belongs in an equally informed portfolio. The complete-graph example above illustrates scope, not newness. The substantive acquisition result here combines unknown sparse *off-diagonal* corruption with low rank and an unknown diagonal, while preserving a sparse residual.

Why deleting to low rank is inadequate. Take a perfect-matching adjacency matrix S and an alternating D of rank r. Then $\mathrm{rank}(D+S) \geq n-r$. Deleting s coordinates can reduce rank by at most $2s$, so making the raw retained matrix rank at most r requires $s \geq (n-2r)/2$. Even free diagonal completion does not cure the example: each matching block with any diagonal has rank at least one, so $\min_v \mathrm{rank}(D+S+\mathrm{diag}\,v) \geq n/2-r$. The present compiler keeps the sparse residual and counts it exactly, avoiding that linear deletion or completion rank.

What failed, and what repaired it. Searching for an inner coefficient matrix costs $2^{\Theta(r^2)}$ and loses large sublinear ranks; distance pruning removes that search. Requiring an alternating low-rank component excludes useful diagonal completions; the fourth-root refinement incorporates them. Requiring pivot coordinates to survive, or dropping the unary substitution correction, gives incorrect algorithms. Executable audits include these counterexamples.

These repairs do not remove the general dense quadratic obstruction. A separated-variable representation of $(-1)^{x^\top By}$ has rational matrix rank $2^{\mathrm{rank}_{\mathbb{F}_2} B}$, so generic full cross-rank requires exponentially many terms in that architecture. Eliminating a Gaussian block can also densify nonlinear interactions:

$$\sum_{u,v\in\mathbb{F}_2^t} (-1)^{u^\top Av + u^\top f(x) + v^\top g(x)} = 2^t (-1)^{g(x)^\top A^{-1} f(x)}$$

for invertible A. Local quadratic f, g can yield $\Theta(t^2)$ quartic terms. These are limitations of the examined mechanisms, not lower bounds on all algorithms. Pure quadratic sums, without a nonlinear or non-Clifford local foreground, are already handled by Gaussian elimination.

Executed finite evidence. The root program `diagonal_completion_compiler.py` receives only the raw matrix and search budgets. Hidden matrices generate fixtures but are not passed to search. It passed 16,933 exact fourth-root Gaussian checks, 5,088 refinement identities, and nine acquired mixed counts at phase orders 2, 4, and 8. The raw searches produced:

n	raw rank	deleted	completion rank	residual edges	seeds tested
8	8	1	1	0	11
12	8	1	1	3	99
14	10	1	2	1	465

Independent checks additionally include eight complete raw full-rank searches through $n = 32$, 36 exact phase sums, 1,659 refinement identities, and 48 radical/magnitude checks of Gaussian sums in dimensions 24, 40, and 64. The alternating implementation and separate validators exercise concentrated error stars, short-word puncturing, proper column supercodes, erased external pivots, empty codes, and invalid-certificate rejection.

The known $O(n^4)$ diagonal-completion algorithm was also implemented and checked against exact optimum for all 1,100 simple graphs through five vertices. On the three raw matrices in the table it returns completion ranks 2, 5, and 6, without sparse residuals or deletions. These are different representations, not a speed comparison. The raw program compiler passed eight exact preimage-count checks, including a 10,001-instruction program with two original inputs and one AND, producing nine phase variables.

The finite inner backend enumerates assignments. These executions validate acquisition and exact reconstruction; they neither execute the asymptotic phase solver nor show a practical speed advantage. The asymptotic result is the proved composition (9), conditional on the specified source results.

9 A strict family with exponential balanced-cut rank

The larger rank range is not merely a redundant way to describe a smaller-rank or small-width family.

Lemma 9.1 (Support cover with a free diagonal)**.** *If a symmetric binary matrix T has rank q and at most W off-diagonal edges, one can find E with $T[R,R] = 0$, $R = [n] \setminus E$, and*

$$|E| \le \left\lfloor \frac{2W}{\tau} \right\rfloor + q\tau \quad \text{for every integer } \tau \ge 1.$$

In particular $|E| \le 2\sqrt{2qW} + q$ for $q > 0$.

Proof. Delete vertices of off-diagonal degree at least τ. Every remaining row has support size at most τ, including its possible diagonal entry. Choose actual rows forming a basis, and delete the union of their supports. It contains every nonzero column, since every row is a linear combination of these basis rows. There are at most q basis rows. Optimize with $\tau = \max(1, \lceil\sqrt{2W/q}\rceil)$; if $q = 0$, the matrix is zero. □

Proposition 9.2 (A family outside the simpler conditions)**.** *Fix $0 < \alpha < 1$ and let $r = 2\lfloor n^\alpha/2 \rfloor$. There are Boolean cubic polynomials $p_n = P_n + q_{B_n}$ with $n/3$ disjoint cubic monomials in P_n, and raw quadratic descriptions of $\Theta(n^2)$ entries, such that:*

1. *$B_n = A_n + S_n$, with A_n alternating of rank r and $e(S_n) = O(n)$, so Theorem 1.2 applies without receiving this decomposition.*
2. *B_n has $\Omega(n^2)$ edges and every balanced original-coordinate cut has binary cut rank $\Omega(n)$. The full phase $(-1)^{p_n}$ has matrix flattening rank $2^{\Omega(n)}$ over $\mathbb{Q}$ or $\mathbb{C}$ across every such cut.*
3. *No symmetric component of rank below r, even with arbitrary diagonal, approximates its off-diagonal matrix within $n^{1+o(1)}$ edge errors. In particular, it cannot be reclassified into the smaller $n^{o(1)}$-rank, near-linear-error regime by choosing another decomposition.*

These are separations between representation conditions, not general counting lower bounds.

Proof. Let J_r be the standard invertible alternating matrix, and choose $U \in \mathbb{F}_2^{n\times r}$ whose column code has minimum distance at least $n/4$ and for which $A = UJ_rU^\top$ has $\Omega(n^2)$ edges. Such a choice exists: for independent uniformly random rows of U, the distance fails with probability at

most $(2^r-1)e^{-n/8}=o(1)$. To check density without assuming independent matrix entries, write $t=2^{-r}$, $N=\binom{n}{2}$, and $W=e(A)$. A direct covariance calculation gives

$$\mathbb{E}W=N(1-t)/2, \qquad \operatorname{Var}W=N(1-t^2)/4+\tfrac{3}{2}\binom{n}{3}t(1-t).$$

Chebyshev's inequality gives $\Pr(W<N/4)=O(n^{-2}+t/n)=o(1)$. Both desired events therefore hold simultaneously. Positive code distance implies $\operatorname{rank}A=r$.

Take S from a simple constant-degree expander family, with maximum degree D and positive constant edge expansion h; such explicit families are available from [7]. Loops can be discarded and bounded parallel multiplicities collapsed without losing constant expansion. Work at sizes divisible by three. Choose a partition into triples with only $O(1)$ edges of S internal to triples: a uniform random partition has expected internal count $2e(S)/(n-1)=O(1)$, so such a partition exists. Let P be the sum of the triples' cubic monomials.

For any cut with both parts of size at least $n/3$, there are at least $hn/3$ crossing expander edges. Discard the $O(1)$ edges within triples. Greedily choose an induced crossing matching, never using two vertices from the same triple. Selecting an edge invalidates at most $O(D^2+D)$ remaining edges, so the matching has $\ell=\Omega(n)$ edges. Its cross submatrix of S is I_ℓ; hence the corresponding submatrix of B has rank at least $\ell-\operatorname{rank}A=\Omega(n)$.

In the full phase flattening, fix every variable outside these matching endpoints to zero. All cubic monomials vanish because each triple retains at most one variable. Quadratic terms confined to a side only multiply rows or columns by signs. The remaining cross character matrix has rank $2^{\operatorname{rank}(I_\ell+A_{\text{cross}})}\geq 2^{\ell-r}$. For completeness, a character matrix $((-1)^{x^\top Hy})_{x,y}$ has exactly $2^{\operatorname{rank}H}$ distinct orthogonal row characters. This proves the claimed flattening rank. Adding $O(n)$ sparse edges to dense A preserves $\Omega(n^2)$ edge density.

Finally suppose a symmetric D' has rank $r'<r$ and off-diagonal error $M'=n^{1+o(1)}$ from B. Then $T=A+D'$ has rank at most $r+r'$ and at most $e(S)+M'$ off-diagonal edges. Lemma 9.1 deletes only $O(\sqrt{r(e(S)+M')}+r)=o(n)$ coordinates to make $T[R,R]=0$. Removing fewer than the code distance leaves U_R of full column rank, so $\operatorname{rank}A[R,R]=r$. But $A[R,R]=D'[R,R]$ has rank at most $r'<r$, a contradiction. □

The cubic third-derivative tensor in this family is a direct sum of three-dimensional volume forms. Its primitive centroid blocks are those triples: a centroid map cannot mix two blocks by nondegeneracy, and the centroid of one volume form is scalar. There are $\Omega(n^2)$ nonzero quadratic couplings between the blocks, since each block pair contains at most nine edges. Within-block changes of basis cannot turn a nonzero cross matrix into zero. Thus the preceding acquired-cubic-block method's near-linear coupling criterion also fails here.

The support-cover proof was checked on every symmetric matrix through dimension five: 168,146 threshold checks. Separate programs checked 2,799 finite balanced cuts, 2,211 restricted full-phase character-rank identities, and all 8,192 rank-zero/rank-one symmetric competitors for a finite rank-two fixture. A larger exact certificate has $n=4,098$, core rank six, code distance 1,993, 16,377 sparse edges, and 4,146,143 observed edges. Any rank-below-six symmetric competitor with at most $4n$ edge errors would imply a support cover of at most 1,698 vertices, contradicting that distance. The larger sampled sparse graph is not claimed to be an expander; its certificate concerns rigidity only.

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